

Experiment# 04

Implementation of Conditional Statements and Loops in Linux

Objective

The objectives of this lab are to familiarize students with shell programming and practice few problems related to

- If-else statements
- For loop
- While loop

in LINUX Ubuntu.

Software Tools

- Ubuntu 16.04
- GCC Compiler
- https://www.tutorialspoint.com/execute_bash_online.php

Description:

This lab has been designed to get even better insight of the shell programming. By the end of this lab you'll be able to solve more complex problems in Linux using shell.

The details of few programming constructs to be used in the tasks are as follows:-

Mathematical Operator in Shell Script	Meaning	Normal Arithmetical / Mathematical Statements	But in Shell	
			For test statement with if command	For [expr] statement with if command
-eq	Is equal to	$a=b$	if test \$a -eq \$b if test \$a -eq \$b;	if [[\$a -eq \$b]] if [\$a -eq \$b];
-ne	Not equal to	$a!=b$	if test \$a -ne \$b if test \$a -ne \$b;	if [[\$a -ne \$b]] if [\$a -ne \$b];
-lt	Less than	$a<b$	if test \$a -lt \$b if test \$a -lt \$b;	if [[\$a -lt \$b]] if [\$a -lt \$b];
-le	Less than or equal to	$a<=b$	if test \$a -le \$b if test \$a -le \$b;	if [[\$a -le \$b]] if [\$a -le \$b];
-gt	Greater than	$a>b$	if test \$a -gt \$b if test \$a -gt \$b;	if [[\$a -gt \$b]] if [\$a -gt \$b];
-ge	Greater than or equal to	$a>=b$	if test \$a -ge \$b if test \$a -ge \$b;	if [[\$a -ge \$b]] if [\$a -ge \$b];

if statements :-

if condition then

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commands

fi

Example

Following script determine whether given argument number is positive.

```
if test $1 -gt 0 then
```

```
echo "$1 number is positive" fi
```

Run it as follows:

```
$ chmod 755
```

```
ispostive ./ ispostive
```

```
5
```

5 number is positive

```
$/ispostive -45
```

Nothing is printed

if else statements:-

```
ifcondition then
```

```
commands
```

```
else
```

```
commands
```

```
fi
```

Example

Following script determine whether given argument number is positive.

```
$ cat>ispostive
```

```
#
```

```
#
```

```
# Script to see whether argument is positive
```

```
#
```

```
number=10
```

```
if [ $number -eq 10 ]; then
```

```
echo "Number is 10."
```

```
else
```

```
echo "Number is not 10."
```

```
fi
```

Run it as follows:

```
$ chmod 755
```

```
ispostive ./ ispostive
```

```
5
```

Number is 10

Case...esac statement in shell:

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These are the statements in shell which replicates the switch case statements in c/c++ language. Syntax is as follows:-

```
#!/bin/bash

# Example variable
option="B"

case "$option" in
    "A")
        echo "Option A selected"
        # Add commands to execute when option A is selected
        ;;
    "B")
        echo "Option B selected"
        # Add commands to execute when option B is selected
        ;;
    "C")
        echo "Option C selected"
        # Add commands to execute when option C is selected
        ;;
    *)
        echo "Invalid option"
        # Add commands to execute when none of the specified options match
        ;;
esac
```

While loop:-

```
while [ condition ]
do
    command1
    command2
    ..
    ....
    commandN
done
```

Example

```
count=1

# While loop

while [ $count -le 5 ]; do
```

```
echo "Count: $count"

count=$((count + 1))

done
```

Lab Tasks

1. **Simulate** suitable command/commands in gedit to input 2 numbers from user and display the greater number.
2. **Simulate** suitable command/commands in gedit to input 2 numbers from user and make a simple calculator which should be capable of performing basic calculations i.e. addition, subtraction, multiplication, division by using if-else conditions. Display all possible outputs.
3. **Simulate** suitable command/commands in gedit to input 2 numbers from user and make a simple calculator which should be capable of performing basic calculations i.e. addition, subtraction, multiplication, division by using switch-case statements. Display all possible outputs.
4. **Simulate** suitable command/commands in gedit to print table of a user defined number.
5. **Simulate** suitable command/commands in gedit to check whether the user defined number is even or odd.

Conclusion:

Lab Instructor

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